

MATH503 Mathematics for Computing

School of Engineering, Computer and Mathematical Sciences

Course Information

Aim

To provide an introduction to the mathematical and statistical concepts required for the analysis of data and for applications in computing.

Content

Sets, probability, probability distributions, statistical measures, functions, relations, matrix operations.

Lectures

You are expected to attend your two-hour face-to-face lecture.

Tutorials

See timetable information on Arion. Part of the credit for the course will be achieved through group work in the tutorials and quizzes.

Office Hours

Students are welcome to discuss problems regarding the course or other matters with their lecturer. Office hours set aside each week specifically for student questions will be announced on Canvas.

Canvas and Online Resources

Students should regularly check the course website on Canvas. This website contains class announcements, assessment and tutorial information, class resources, video recordings to watch prior to lectures and tutorials, as well as assessment marks. The resources on Canvas are integral to the course. Please familiarise yourself with these.

Myimaths

www.myimaths.com is used for your assignment. Personal login details will be supplied by Week 2. In the meantime, you are welcome to use the school username and password to familiarise yourself with the site.

School username: computer

School password: tangent179

Academic support for students

Additional academic support is available at AUT. Look for information on *ECMS study space* on Canvas.

Textbook

Recommended textbooks for Sets, Counting Techniques and Probability are available in the High Demand sections of the library:

[LGR] *Finite mathematics* by Margaret Lial, Raymond Greenwell, and Nathan Ritchey

[T] *Finite mathematics for the managerial, life, and social sciences* by Soo Tan

Extracts from these books are available on the *Course Resources link* on Canvas.

Additional reading will be advised on Canvas.

Collaboration

Students are welcome to discuss their assessments with their lecturers, tutors and with other students. However, unless otherwise stated, all the student's work must be their own. Hence no student may receive any part of an individual assessment from another person, whether in printed or electronic form. Failure to abide by this will result in a possible academic integrity investigation.

So that you are fully aware of your obligations around academic integrity, please complete the modules on academic integrity of Canvas.

Assessment

The assessment consists of assignments and a comprehensive problem solving questionnaire.

Assessment Event	Weight	Details about the assessment
Portfolio	30%	Group tasks in tutorials and quizzes Ongoing
Assignment	25%	Myimaths online activities Ongoing
Final Controlled Assessment	45%	3 hours in Week 13 to 15 (date to be announced)

Students are expected to attempt all assessments.

In order to pass the course, a student should gain a minimum mark of 50% overall.

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Chapter 1

Sets and probability

1.1 Probability

Calculating probabilities

Probability deals with uncertainty in the real world.

Example 1: Theoretical probability

A die is fair if any one of the six faces is equally likely to come up when the die is rolled. A single fair die is rolled once.

Find the probability for

- getting a 2
- getting an odd number
- getting a number greater than 2

In this case we assume that the six possible outcomes of rolling the die are all *equally likely*. The probability is calculated as

$$\text{Probability} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$$

Example 2: Empirical probability

The following table lists the preferred programming language of a group of 2000 IT professionals. The probability that a randomly selected person from this group has a particular preference is calculated as the *relative frequency*. Clearly, the preferred programming languages are not equally likely.

Preferred language	Number of IT professionals	Empirical probability
Javascript	820	$\frac{820}{2000} = 0.41$
Python	240	
C++	60	
Java	880	
Total	2000	

These empirical probabilities should have a total of 1. If the sampled group is large enough, an empirical probability gives a good approximation for the probability that any randomly selected IT professional has a particular preference.

Example 3: Subjective probability

What is the probability that the All Blacks will win their next game?

Subjective probability is an individual person's measure of belief that an event will occur. It is expressed as a number from 0 to 1, or as a percentage from 0% to 100%. In this paper we will not consider subjective probability.

Some problems from the early days of probability theory

- How many times do you have to roll a die so that the probability for getting at least one six is more than 50%?
- *Problems of points*: Two players agree to play a series of contests. They have equal chances of winning each round. The one who first reaches 3 wins is the victor and takes home the prize. They have to abandon their contest when one player leads by 2 games to 1. How should they split the prize fairly?

These problems were studied (and solved) by Fermat and Pascal in 1654.

1.2 Sets

Reading: [T] p324 – 330 or [LGR] p342 – 349

Definition and notation

A **set** is any collection of objects. The objects are called the **elements** of the set.

We write $a \in A$ and say that a is an element of the set A .

If b is not an element of A , we write $b \notin A$.

Sets are often defined by listing the elements between the braces $\{ \}$, such as the set of colours of the rainbow:

$$A = \{\text{red, orange, yellow, green, blue, indigo, violet}\},$$

or the set of all **natural numbers** (the positive integers):

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

Set builder notation

If it is impractical to list the elements of a set it is often defined by specifying a property $p(x)$ for the elements:

$$A = \{x : p(x)\}.$$

The set A consists of all those elements x for which $p(x)$ is true.

The colon $:$ is read as *such that*

Cardinality

The **cardinality** $n(A)$ of a set A is the number of distinct elements that it contains. Another notation for cardinality is $|A|$.

Some sets have an infinite number of elements.

Example 1

$$A = \{x : x \text{ is a colour of the rainbow}\} = \{\text{red, orange, yellow, green, blue, indigo, violet}\}$$

$$n(A) =$$

$$B = \{x : x \text{ is a natural number less than 5}\} =$$

$$n(B) =$$

$$C = \{x: x \text{ is an integer greater or equal to } -2 \text{ and less than } 4\} =$$

$$n(C) =$$

Empty set

The **empty set** is denoted by $\{\}$ or by $\emptyset$. The empty set contains no elements.

Example 2

Find the cardinality of the following sets. Note that a set can have sets as elements.

$$n(\emptyset) =$$

$$A = \{1, \{1, 2\}\}, n(A) =$$

$$B = \{\emptyset, \{1, 2\}, \{1, 3\}, \{2, 3\}\}, n(B) =$$

$$C = \{\emptyset, \{\emptyset\}\}, n(C) =$$

Equal sets

Two sets A and B are said to be **equal** if they have precisely the same elements. The order in which the elements appear is not important.

Example 3

Show that the two sets A and B are equal.

$$A = \{x: x \text{ is a prime number less than } 5\} =$$

$$B = \{x: x^2 - 5x + 6 = 0\}$$

$$\text{Solve } x^2 - 5x + 6 = 0$$

$$A = B$$

Universal set

The **universal set** is the set consisting of all the possible elements under consideration.

Example 4

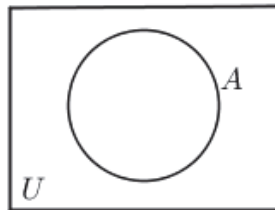
If the universal set is all positive integers, then $\{x: x^2 = 4\} = \{2\}$.

If the universal set is all integers, then $\{x: x^2 = 4\} = \{-2, 2\}$.

Venn diagrams

A **Venn diagram** gives a visual representation of sets. Usually the universal set is denoted by a rectangle, and all elements are considered as points inside the rectangle.

A set of some of the elements is drawn as a circle around the elements which it contains.



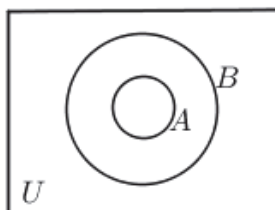
Subsets

A set A is said to be a **subset** of another set B , denoted by $A \subseteq B$, if every element that is in the set A is also in the set B .

A **proper subset** A of B is a subset of B that is not equal to B and we write $A \subset B$. There exists at least one element of B that is not in A .

On a Venn diagram if $A \subset B$ then the circle representing the set A is drawn inside the circle representing the set B .

For any set A , $\emptyset \subseteq A$ and $A \subseteq A$.



Example 5

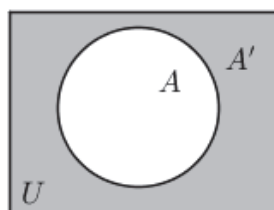
Find all the subsets of $A = \{1, 2, 3\}$. Which of these subsets are proper subsets?

Complement of a set

The **complement** of a set A is the set A' consisting of all elements in the universal set which do not belong to A , so

$$A' = \{x : x \notin A \text{ and } x \in U\}.$$

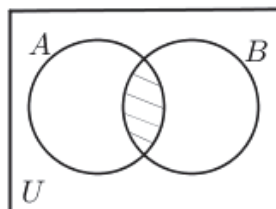
This is represented on a Venn diagram as the region outside of the circle representing the set A . Some textbooks use the notation $\bar{A}$ or A^c for the complement of set A .



Intersection of sets

The **intersection** of two sets A and B is the set $A \cap B$ consisting of all elements which belong to both A and B , so

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$



Disjoint sets

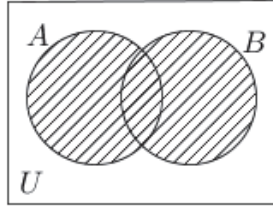
Two sets A and B are said to be *disjoint* if they have no elements in common, so $A \cap B = \emptyset$. This is represented on a Venn diagram as two regions that don't intersect.



Union of sets

The **union** of two sets A and B is the set $A \cup B$ consisting of all elements which belong to A or to B or both, so

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$



Example 6

Let $A = \{1, 3, 5, 7, 9\}$, $B = \{3, 6, 9\}$, $C = \{1, 2, 3, 4, 5\}$ and the universal set $U = \{x : x \text{ is a positive integer and } x \leq 10\}$. Find the following sets:

- $A \cup B =$
- $A' \cap B' =$
- $A \cap (B \cup C) =$
- $(A \cap B) \cup (A \cap C) =$
- $A \cup B' =$
- $\emptyset' =$
- $U' =$

Example 7

Let $B = \{a, b, c, \{d\}, \{e, f\}\}$. Answer each of the following as *true* or *false*.

- $a \in B$
- $\{b, c, d\} \subset B$
- $\{d\} \in B$
- $\{d\} \subseteq B$
- $\{a, \{e, f\}\} \subset B$

Exercises

1. List the elements of the following sets:
 - (a) The set A consisting of the four seasons.
 - (b) $B = \{n : n \text{ is an even positive integer and } n < 8\}$
2. Given that $X = \{1, 2, 3, 4\}$, list all the subsets A of the set X that satisfy the given conditions:
 - (a) $A \subseteq X$ and $n(A) = 2$
 - (b) A is a proper subset of X
 - (c) $A \subseteq X$ and $1 \in A$

1.3 Applications of Venn diagrams

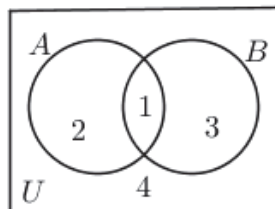
Reading: [T] p335 – 336 or [LGR] p353 – 361

Survey problems

In a survey problem, data is collected about properties of individuals or objects. (See Example 1 below.) In the language of sets, we have information about the cardinality of certain sets and some of their unions and intersections. This information can then be used to determine some other facts about the situation. Summarizing the given data on a Venn diagram is often a useful technique to visualize the data.

Two sets

In a Venn diagram for two sets A and B , the universal set U is divided up into 4 regions. Number the regions and express each region in terms of an intersection of A , B , A' and B' .

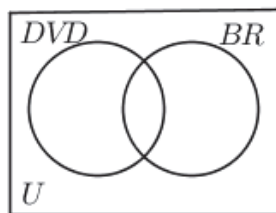


Example 1

In a survey of 100 households it is found that 76 households have a DVD player, 21 have a Blu-Ray player and 12 have both.

How many households have either a Blu-Ray player or a DVD player or both? How many have neither a DVD player nor a Blu-Ray player? How many have a DVD player but not a Blu-Ray player?

Solution

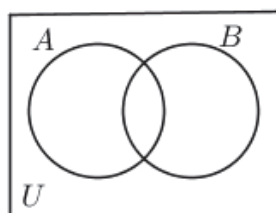


Union rule for sets

If A and B are finite sets, then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

We use the following Venn diagram to prove the rule:



Proof

When A and B are disjoint, the union rule simplifies to

$$n(A \cup B) = n(A) + n(B)$$

Explain why.

Example 2

In a survey of 100 households it is found that 76 households have a DVD player, 21 have a Blu-Ray player and 12 have both.

How many households have either a Blu-Ray player or a DVD player or both?

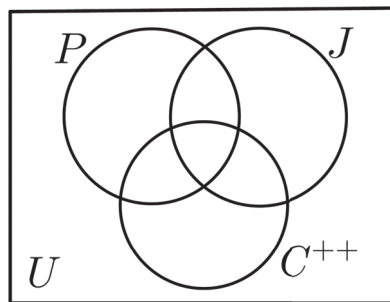
Solution

Three sets

Example 3

In a survey of 100 students regarding their programming experience the following information is obtained: 40 has experience in using Python, 55 in using Java, and 35 in using C⁺⁺. Moreover, 25 have experience in using both Python and Java, 5 in using Java and C⁺⁺, and 15 in using Python and C⁺⁺. Only 5 people have experience in using all three languages. How many students in the survey have no experience in using these three languages?

Solution



Exercises

- Sketch a Venn diagram for the universal set U and two intersecting sets A and B . Shade the regions representing the following sets:
 - $B \cap A'$
 - $A \cup B'$
 - $B' \cup (A' \cap B')$
- In a survey in a shopping center on people's cooking facilities the following results were obtained: 58 households use microwave ovens, 63 use electric ranges, 58 use gas ranges, 19 use microwave ovens and electric ranges, 17 use microwave ovens and gas ranges, 4 use both gas and electric ranges, 1 uses all three and 2 used none of the three.
How many people or households were interviewed?

1.4 Probability and sets

Reading: [T] p364 – 369, p372 – 376 or [LGR] p365 – 371

Sample spaces

In probability we study situations that involve random phenomena.

An **experiment** is an activity with an observable result.

Each repetition of an experiment is called a **trial**.

The result of a trial is called an **outcome**.

The **sample space** for an experiment is the set of all possible outcomes.

Example 1

Experiment 1 Toss a coin once to get head or tails.

The sample space is $S = \{h, t\}$.

Experiment 2: In a study the number of boys and girls in families with exactly three children are determined.

If the order of the births are not considered, the sample space is

$$S_1 =$$

If the order of the births are recorded, the sample space is

$$S_2 =$$

Experiment 3: Toss a coin until both a head and a tail appear, or until the coin has been tossed four times.

The sample space is

$$S =$$

Equally likely outcomes

Experiment 2 shows that there may be more than one sample space for a specific experiment. The four outcomes in the sample space S_1 are *not equally likely*. The outcomes in sample space S_2 are *equally likely* if we assume that boys and girls are equally likely. We prefer to work with a sample space with equally likely outcomes.

Events

An **event** is a subset of the sample space.

Example 2

Experiment 2 Use sample space S_2 .

Write the following event in set notation.

E : The family has exactly two daughters.

$E =$

Write the following event in words and in set notation.

E' :

$E' =$

Experiment 3 Write the following events in set notation.

G : The coin is tossed exactly 3 times.

$G =$

H : The coin is tossed at least 3 times.

$H =$

Mutually exclusive events

Events E and F are **mutually exclusive** events if $E \cap F = \emptyset$.

Events E and E' are always mutually exclusive.

Example 3

Use sample space S_2 for Experiment 2.

Event E : the family has exactly 2 daughters.

Event F : the family has exactly 1 daughter.

$E \cap F =$

Probability

Let S be a sample space with **equally likely** outcomes, and let event E be a subset of S .

Then the **probability** that event E occurs is

$$P(E) = \frac{n(E)}{n(S)}.$$

Property of probability

For any event E , $0 \leq P(E) \leq 1$.

Proof

Union rule

For any events E and F from a sample set S ,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

Proof:

Example 4 ([LGR] p377)

Suppose two fair dice are rolled. Find the probability that the first die shows a 2, or the sum of the results is 6 or 7.

Solution

Sample space S :

1-1	1-2	1-3	1-4	1-5	1-6
2-1	2-2	2-3	2-4	2-5	2-6
3-1	3-2	3-3	3-4	3-5	3-6
4-1	4-2	4-3	4-4	4-5	4-6
5-1	5-2	5-3	5-4	5-5	5-6
6-1	6-2	6-3	6-4	6-5	6-6

Event A is *the first die shows a 2*

Event B is *the sum of the results is 6 or 7*

Event $A \cup B$ is *the first die shows a 2, or the sum of the results is 6 or 7*

$$P(A) = \frac{n(A)}{n(S)} =$$

$$P(B) = \frac{n(B)}{n(S)} =$$

$$P(A \cap B) =$$

$$P(A \cup B) =$$

Complement rule

For any event E from a sample space S ,

$P(E) = 1 - P(E') \quad \text{and} \quad P(E') = 1 - P(E).$

Proof:

Example 5:

Suppose two fair dice are rolled. Find the probability that the sum of the numbers rolled is greater than 3.

Solution:

Odds

The **odds in favour** of an event E are defined as

$$\frac{P(E)}{P(E')}.$$

For this definition, $P(E') \neq 0$.

If the odds favouring event E are m to n , then

$$P(E) = \frac{m}{m+n} \quad \text{and} \quad P(E') = \frac{n}{m+n}.$$

Example 6

The odds for rain tomorrow are 4:7. What is the probability that it will rain tomorrow? What is the probability that it will not rain tomorrow?

Solution

Example 7

It is known that 13% of people prefer to manipulate their computer mouse with their left hand. What are the odds that a randomly chosen person prefers to manipulate their mouse not with their left hand?

Solution

Probability distributions – Chapter 4

For a sample space S with n distinct outcomes $s_1, s_2, \dots, s_n$, which are not equally likely, a probability p_i is assigned for each outcome s_i . Then

$$0 \leq p_i \leq 1 \quad \text{for each } i$$

and

$$p_1 + p_2 + \dots + p_n = 1.$$

Exercises

1. List the elements of the following sample space:
 - (a) S is the set of all outcomes of rolling two distinguishable dice such that the numbers add up to 8.
 - (b) S is the set of all outcomes of rolling two indistinguishable dice such that the numbers add up to 8.
2. An unprepared student takes a three-question true/false quiz in which she guesses the answers to all three questions.
 - (a) Write out the sample space S for her three answers, recording whether she got the answers correct or incorrect.
 - (b) What is $n(S)$?
 - (c) Write the event *The student gets all three answers correct* in set notation.
 - (d) Write the event *The student gets exactly two answers correct* in set notation.
 - (e) Write the event *The student gets only the first answer correct* in set notation.
 - (f) Calculate the probabilities for the events.
3. A card is drawn from a well-shuffled deck of 52 cards. Find the probability of drawing the following:
 - (a) a heart
 - (b) a black card
 - (c) a 2 or a queen
 - (d) a face card
 - (e) a black 7 or a red 8
 - (f) a spade or a king
4. Decide whether the following are mutually exclusive events:
 - (a) Owning a dog and owning an iPhone.
 - (b) Being retired and being 70 years old.
 - (c) Being a teenager and being 70 years old.

1.5 Conditional probability

Reading: [T] p398 – 408 or [LGR] p390 – 400

Conditional probability, Venn diagrams and tables

The **conditional probability** of event E , given event F is

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \text{ where } P(F) \neq 0.$$

Example 1

There are 200 students in a school of which 80 are female students. Of the 200 students, 140 are studying astronomy of which 40 are female. Let F be the event that a student is female and A the event that a student studies astronomy.

(a) Sketch a Venn diagram for the problem.

(b) Complete the following table:

	Female	Male	Total
Study astronomy			
Do not study astronomy			
Total			200

(c) What is the probability that a student is studying astronomy?

$$P(A) =$$

(d) If the student is female, what is the probability that she is studying astronomy? Use the Venn diagram or the table to calculate this probability.

$$P(A|F) =$$

(e) Use the formula for conditional probability to find the probability that the student studies astronomy, given that the student is female.

$$P(A|F) = \frac{P(A \cap F)}{P(F)}$$

$$P(F) =$$

$F \cap A$: student studies astronomy and is female

$$P(A \cap F) =$$

$$P(A|F) =$$

(f) Find $P(A|F')$ and explain the meaning of this conditional probability.

$$P(A \cap F') =$$

$$P(F') =$$

$$P(A|F') = \frac{P(A \cap F')}{P(F')} =$$

$P(A|F')$ is the probability that a student studies astronomy, given that the student is male.

Independent events

Events E and F are **independent events** if

$$P(F|E) = P(F) \text{ or } P(E|F) = P(E).$$

Example 2

Are the events F and A in Example 1 independent events?

Solution

Product rule of probability

If E and F are events, then $P(E \cap F)$ may be found by a product formula.

$$P(E \cap F) = P(F)P(E|F) \quad \text{or} \quad P(E \cap F) = P(E)P(F|E).$$

This rule is often used where an experiment is done as a sequence of two steps, where E is the event on the first step and F the event on the second step.

Example 3

A box contains 6 good items and 2 defective items. Two items are selected at random from the box without replacement. Find the probability that both selected items are good.

Solution

E : the item selected first is good

F : the item selected second is good

We have to find $P(E \cap F)$.

$P(E) = \frac{6}{8}$. Explain why.

What is the sample space S for this part of the experiment?

$P(F|E) = \frac{5}{7}$. Explain why.

What is the sample space S for this part of the experiment?

$$P(E \cap F) = P(E)P(F|E) = \frac{6}{8} \times \frac{5}{7} = \frac{15}{28}.$$

Product rule and tree diagrams

Tree diagrams are used with the product rule to find the probability for a sequence of events.

Example 4

Present the calculation in Example 3 on a tree diagram.

Solution**Product rule for independent events**

Events E and F are independent events if and only if

$$P(E \cap F) = P(E)P(F).$$

Example 5

It is known that the failure of the hard drive of a computer is independent of the failure of the sound card. The probability of a hard drive failure on any given day is 0.00015, and the probability of a sound card failure 0.00025.

What is the probability on a given day of both a sound card and a hard drive failure?

Solution**Exercises**

1. Two cards are drawn from a standard deck, one after the other without replacement. Find the probability that
 - (a) the second is a face card, given that the first card is a jack.
 - (b) the second is a an ace, given that the first card is not an ace.
 - (c) Use the product rule to find the probability that the first card is a heart and the second card is red.

2. The probability that a new medicine is effective is 0.75 and the probability that a patient will have a certain side effect is 0.4. The probability that the medicine is effective and the patient experiences the side effect is 0.3. Decide whether these events are dependent or independent.
3. Let A and B be independent events with $P(A) = 0.25$ and $P(B) = 0.2$. Find $P(A \cap B)$ and $P(A \cup B)$.
4. If A and B are events such that $P(A) = 0.5$ and $P(A \cup B) = 0.7$, find $P(B)$ when
 - (a) A and B are mutually exclusive;
 - (b) A and B are independent.
5. Use a tree diagram to solve the following problem.

In a certain area, 15% of the population are joggers and 40% of the joggers are women. If 55% of those who do not jog are women, find the probabilities that an individual from the community fits the following description:

- (a) a woman jogger
- (b) not a jogger
- (c) a woman

Chapter 2

Counting principles and probability

2.1 Permutations

Reading: [T] p340 – 350 or [LGR] p432 – 439

Probability

Let S be a sample space of equally likely outcomes, and E an event.

The probability that event E occurs is $P(E) = \frac{n(E)}{n(S)}$.

When an experiment has many outcomes, that is $n(S)$ is large, it is not possible to list all the outcomes in S and counting principles are needed to find $n(S)$ and $n(E)$.

Multiplication principle

Suppose n choices are to be made with

m_1 ways to make choice 1,

and for each of these choices,

m_2 ways to make choice 2,

and so on, with

m_n ways to make choice n .

Then there are $m_1 \times m_2 \times \cdots \times m_n$

different ways to make the sequence of choices.

Example 1

A combination lock can be set to open to any 3-letter sequence. How many sequences are possible?

How many sequences are possible if no letter is repeated?

Solution**Example 2**

A computer word is made up of 8 bits, each bit is a digit 0 or 1. How many different computer words of this type is possible?

Solution**Factorial notation**

For any natural number (positive integer) n ,

$$n! = n \times (n - 1) \times (n - 2) \cdots \times 3 \times 2 \times 1 \quad \text{and} \quad 0! = 1.$$

Example 3

Simplify the following expressions:

- $1! =$
- $2! =$
- $3! =$
- $5! - 4! =$

- $\frac{6!}{3!} =$
- $\frac{6!}{2!3!} =$
- $\frac{n!}{(n-2)!} =$
- $\frac{(n-2)!}{(n-1)!} =$
- $\frac{n!}{0!} =$
- $n(n-1)! =$
- $n! - (n-1)! =$

Permutations

A **permutation** of r elements from a set of n elements is any specific ordering of the r elements.

The number of permutations of n objects taken r at a time is written $P(n, r)$.

Example 4

In a club with 16 members, how many ways can a committee of 3 officers (president, secretary, treasurer) be chosen?

Solution

A set of 3 members have to be chosen from 16 members.

Each choice will be a permutation.

The order in which the 3 members are chosen is important. The first one is the president, the second one the secretary and the third one the treasurer.

Use the multiplication principle:

First choice:

Second choice:

Third choice:

Number of ways to select the committee:

Formula for permutations

General rule for $0 \leq r \leq n$: $P(n, r) = \frac{n!}{(n-r)!}$

Proof:

Example 5

Repeat Example 4, but use a permutation to calculate the number of ways in which to select the committee.

Solution

Exercises

1. (a) How many 4-digit keys for an electronic lock can be made, if the first digit is a 1 or a 2, and no digits may be repeated?
(b) How many 4-digit keys can be made, if the first digit is a 1 or a 2, and digits may be repeated?
2. Evaluate the following permutations:
 $P(13, 2)$, $P(13, 11)$, $P(n, 1)$, $P(n, n - 1)$, $P(n, n)$.
3. There are 5 people who have to address a convention. In how many different orders can they appear?
4. How many 7-digit telephone numbers are possible if the first digit cannot be a zero, and
 - only odd digits may be used,
 - the first three digits are 481,
 - no repetition of digits are allowed.
5. (a) An operating system has passwords of exactly four lower case letters. How many possible passwords are there?
(b) If the system can distinguish between lower case and upper case letters, there will be twice as many possible passwords. True or false? Why?

2.2 Combinations

Reading: [T] p350 – 354 or [LGR] p442 – 449

Combinations

A **combination** is an unordered arrangement of objects or items.

The number of combinations of r objects chosen from n distinct items are written as $C(n, r)$.

Example 1

How many ways are there to select a committee of 3 people from a group of 16 people?

Solution

First calculate the number of ways in which 3 people can be selected from 16 people in an ordered way.

Use a permutation:

$$P(16, 3) =$$

As the positions on the committee are not specified, the order in which the people is selected should not be taken into account. Each group of 3 will appear a few times in the ordered selections. How many times?

Use a permutation:

$$P(3, 3) =$$

The number of combinations, or unordered selections, of 3 persons from 16 people:

$$C(16, 3) = \frac{P(16, 3)}{P(3, 3)} =$$

Formula for combinations

In general for $0 \leq r \leq n$, the number of combinations of r elements chosen from n elements, is given by

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

Proof**Example 2**([LGR] p445)

A salesman has 10 accounts in a certain city.

- (a) In how many ways can he select 3 accounts to call on?

- (b) In how many ways can he select 7 accounts to call on?

- (c) In how many ways can he select at most 4 accounts to use in preparation of a report?

- (d) The salesman has to visit 3 of the accounts picked at random. How many different traveling circuits are possible?

Example 3

How many eight-digit binary numbers have exactly three 1s in them?

Solution**Counting techniques**

Four different counting techniques may be used. In some cases more than one of the techniques may be used to get the same answer:

- multiplication principle,
- permutations,
- combinations,
- tree diagrams.

Exercises

1. Evaluate the following combinations:
 $C(12, 5)$, $C(44, 20)$, $C(n, 0)$, $C(n, n - 1)$.
2. Show that $C(n, r) = C(n, n - r)$.
3. In a club with 9 male and 11 female members, how many 5-member committees can be chosen with all women? How many 5-member committees can be chosen with 3 men and 2 women? How many 5-member committees with at least 4 women?
4. In how many ways can 6 diamonds be chosen from an ordinary deck of playing cards?
5. For a poker hand, five cards are chosen from an ordinary deck of playing cards. In how many ways is it possible to get a hand with 4 queens? In how many ways is it possible to get a hand with no face cards? In how many ways is it possible to get a hand with at least 2 face cards?

2.3 Counting principles and probability

Reading: [T] p391 – 395 or [LGR] p454 – 462

Summary of principles of probability

Definition

$$P(E) = \frac{n(E)}{n(S)}$$

Union rule

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

E and F are mutually exclusive if $E \cap F = \emptyset$

$P(E \cup F) = P(E) + P(F)$ if E and F are mutually exclusive

Complement rule

$$P(E') = 1 - P(E)$$

Product rule

$$P(E \cap F) = P(F)P(E|F) \text{ or } P(E \cap F) = P(E)P(F|E)$$

E and F are independent if $P(F|E) = P(F)$ or $P(E|F) = P(E)$

$P(E \cap F) = P(E)P(F)$ if E and F are independent

Examples

Example 1 ([LGR] p457)

When shipping diesel engines abroad, 12 engines are packed into a container at the plant to be sent to the port. A spot check is made of the engines after loading. Three randomly selected engines from a container are tested. If any of the 3 engines are defective, the container will not be shipped.

Suppose a given container has 2 defective engines. Find the probability that the container will not be shipped.

Method 1

$$P(\text{container is not shipped}) = P(1 \text{ engine fails test}) + P(2 \text{ engines fail test})$$

In more formal notation:

Event E : container is not shipped

Event E_1 : 1 engine fails test

Event E_2 : 2 engines fail test

Now $E = E_1 \cup E_2$ and

$P(E) = P(E_1) + P(E_2)$ from the union rule, as $E_1 \cap E_2 = \emptyset$.

The sample space S will be all the possible ways in which three engines can be selected from the 12 engines in the container.

Method 2

$$P(\text{container is not shipped}) = 1 - P(\text{container is shipped}) = 1 - P(0 \text{ engines fail test})$$

Method 3

Use a probability tree to calculate $P(0 \text{ engines fail test})$

Exercises

1. An elevator has 4 passengers and stops at 7 floors. Is is equally likely that a person will get off at any one of the 7 floors. Find the probability that at least 2 passengers leave the lift at the same floor.
2. Twenty-six slips of paper are each marked with a different letter of the alphabet and placed in a container. A slip is pulled out, its letter is recorded (in the order in which the slip was drawn) and the slip is returned to the container. This is repeated 5 times. Find the probabilities that the following “words” are formed.
 - (a) Chuck
 - (b) A word that starts with “p”
 - (c) A word with no repetitions of letters
 - (d) A word that contains no “x”, “y” or “z”
3. In a game show you have six envelopes to choose from. Four envelopes are empty and two envelopes contain a \$100 bill each. You are told to choose 2 envelopes at random. Which is more likely: (1) that you will get at least one \$100 bill, or (2) that you will get no \$100 bill? Calculate both the probabilities.
4. A student is late for class and grabs two socks from a drawer containing 9 black, 6 brown and 2 blue socks. What is the probability that she picks a matching pair?

Chapter 3

Statistical Measures

Reading: <https://statistics.laerd.com/statistical-guides/measures-central-tendency-mean-mode-median.php>

3.1 Measures of Central Tendency

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.

The mean, median and mode are all valid measures of central tendency, but under different conditions, some measures of central tendency become more appropriate to use than others.

3.1.1 The mean

The mean is the most popular and well known measure of central tendency.

The mean is equal to the sum of all the values in the data set divided by the number of values in the data set. If we have n values in a data set and they have values $x_1, x_2, \dots, x_n$ the sample mean, usually denoted by $\bar{x}$ is:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

This formula is written in a slightly different manner using the Greek capital letter, Σ , pronounced "sigma", which means "sum of...":

$$\bar{x} = \frac{\sum x_i}{n}$$

The population mean is usually denoted by μ . For a population with N values

$$\mu = \frac{\sum x}{N}$$

In statistics sample and population have different meanings.

If we were looking at this semester's MATH503 students' attendance, the population is the attendance of all students taking this course (332 students). A sample of this population would be looking at the attendance of 50 students taking this course. The sample is only part of the population, hence why we use μ to denote the population mean and $\bar{x}$ to denote the sample mean.

The mean is susceptible to data sets that contain outliers (extreme data values that are away from the main body of data).

For example:

Eleven staff have the following salaries:

45k, 48k, 56k, 54k, 55k, 55k, 46k, 32k, 47k, 120k, 135k

$$\bar{x} = \frac{45 + 48 + 56 + 54 + 55 + 55 + 46 + 32 + 47 + 120 + 135}{11}$$

$$\bar{x} = 63\text{k}$$

The mean salary is 63k. However, inspecting the raw data suggests that this mean value might not be the best way to accurately reflect the typical salary of a worker, as most workers have salaries in the \$45k to \$56k range. The mean is being skewed by the two large salaries. Therefore, in this situation, we would like to have a better measure of central tendency. The median would be a better measure of central tendency in this situation.

3.1.2 The median

The median is the middle score for a set of data that has been arranged in order of magnitude.

To find the median, first order the data in order of magnitude, generally from smallest to largest. Then find the middle score.

- For example: 3, 4, 6, 8, 11.

We can see 6 is the middle value, therefore 6 is the median

3, 4, 6, 8, 11

- 2, 3, 4, 5, 7, 9. For this data the middle value lies between 4 and 5, therefore the median is 4.5 ($\frac{4+5}{2} = 4.5$)

2, 3, 4 | 5, 7, 9

The median is always representative of the middle data and is less affected by extreme data (outliers).

For example:

For the eleven staff's salaries listed in order of magnitude

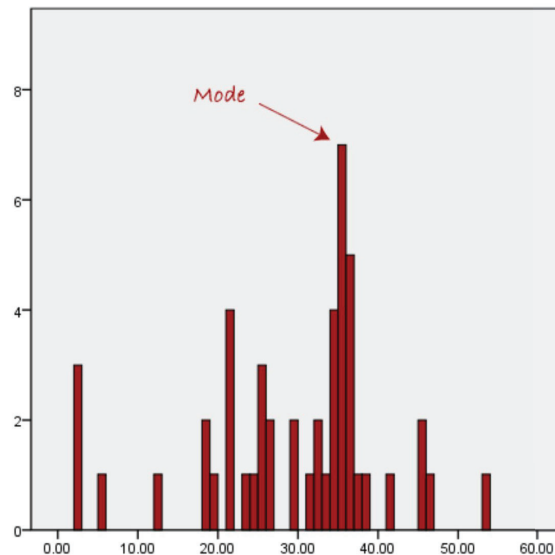
32k, 45k, 46k, 47k, 48k, 54k, 55k, 55k, 56k, 120k, 135k

The median is 54k.

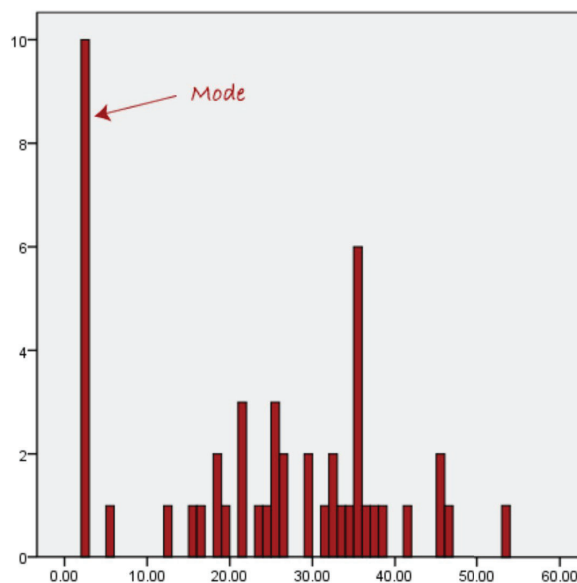
This is far more representative of the middle of the data than the mean which is 63k.

3.1.3 The mode

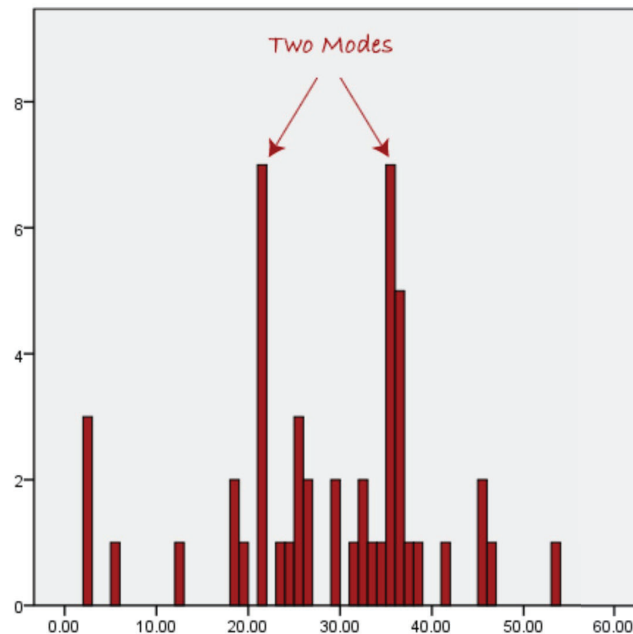
The mode is the most common score from the data set. This is useful if you want to find out what are the most popular scores, though in general the mode is the least useful of the measures of central tendency.



For example: If the most common score is far away for the data set the mode is not a good measure of central tendency. This can be very interesting in itself but does not reflect the centre of the data.



Data can have more than one common score. This makes it difficult to know which mode best describes the central tendency of the data.



3.1.4 Using R studio to calculate measures of central tendency.

It is easy to calculate the mean in R using the `mean()` function and `median()` function:

Mean(dataset)

Median(dataset)

R does not have a standard in-built function to calculate mode. We will need to create a user function in order for R to calculate this for us. We will not be covering this in this course.

https://www.tutorialspoint.com/r/r_mean_median_mode.htm

3.2 Measures of Spread

A measure of spread, sometimes also called a measure of dispersion, is used to give us information about how spread out or how close together a set of data is. Another way of thinking about this is that it tells us about the variability in a sample or population.

A measure of spread is usually used in conjunction with a measure of central tendency, such as the mean or median, to provide an overall description of a set of data.

One use of a measure of spread is that it gives us an idea of how well the measures of central tendency represent the data. For example, if the spread of values in the data set is large, the mean is not as representative of the data as when the spread of data is small. This is because a large spread indicates that there are large differences between individual scores.

We will be looking at the following measures of spread:

range, quartiles, variance and standard deviation.

3.2.1 Range

The range is the difference between the highest and lowest scores in a data set and is the simplest measure of spread.

We calculate range as:

Range = maximum value - minimum value

For example, let us consider our salary data set listed in order of magnitude:

32k, 45k, 46k, 47k, 48k, 54k, 55k, 55k, 56k, 120k, 135k

The maximum value is

The minimum value is

The range is , which is the maximum value minus the minimum value.

Using the range as a measure of spread can be limited, but it does set the boundaries of the scores. This is useful if you are measuring a variable that has either a critical low or high threshold (or both) that should not be crossed. The range will instantly inform you whether at least one value broke these critical thresholds. In addition, the range can be used to detect any errors when entering data. For example, if you have recorded the age of school children in your study and your range is 7 to 123 years old you know you have made a mistake!

3.2.2 Quartiles

Quartiles tell us about the spread of a data set by breaking the data set into quarters, just like the median breaks it in half.

Using our salary data set

32k, 45k, 46k, 47k, 48k, 54k, 55k, 55k, 56k, 120k, 135k

First find the median and remove it. In this case the median is 54k.

The median splits the data in an upper group and a lower group.

Lower group: 32k, 45k, 46k, 47k, 48k,

Upper group: 55k, 55k, 56k, 120k, 135k

Now find the median of the lower group. This is your lower quartile.

Lower quartile =

Next find the median of the upper group. This is your upper quartile.

Upper quartile =

Quartiles are a useful measure of spread because they are much less affected by outliers or a skewed data set than the equivalent measures of mean and standard deviation. For this reason, quartiles are often reported along with the median as the best choice of measure of spread and central tendency, respectively, when dealing with skewed and/or data with outliers.

3.2.3 Interquartile range

A common way of expressing quartiles is as an interquartile range. The interquartile range describes the difference between the upper quartile (Q3) and the lower quartile (Q1), telling us about the range of the middle half of the scores in the distribution.

Hence, for our salary data set:

Q3=

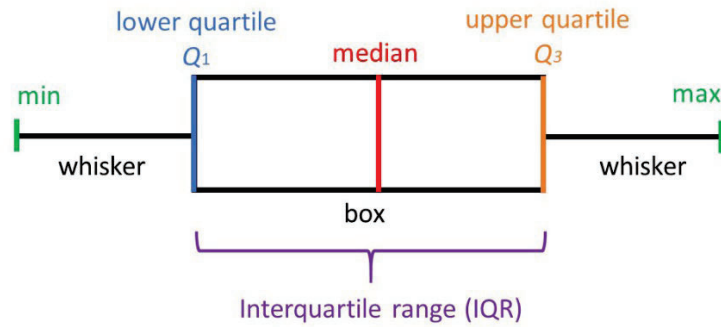
Q1=

Interquartile range = Q3 - Q1

=

3.2.4 Box and Whisker graphs

Box and whisker graphs (also known as box plots) effectively display the range and quartiles.



3.2.5 Variance

Quartiles are useful, but they are also somewhat limited because they do not take into account every score in our group of data. To get a more representative idea of spread we need to take into account the actual values of each score in a data set. The variance is one such measure.

Variance is one way of calculating the deviation of a group of scores from the mean.

$$variance = \frac{\sum(X - \mu)^2}{N}$$

Where μ = mean, $\sum$ = sum of, X = data piece (score), N = number of data pieces (scores)

As a measure of variability, the variance is useful. If the scores in our group of data are spread out, the variance will be a large number. Conversely, if the scores are spread closely around the mean, the variance will be a smaller number. However, there are two potential problems with the variance. First, because the deviations of scores from the mean are 'squared', this gives more weight to extreme scores. If our data contains outliers (in other words, one or a small number of scores that are particularly far away from the mean and perhaps do not represent well our data as a whole), this can give too much weight to these scores. Secondly, the variance is not in the same units as the scores in our data set: variance is measured in the units squared. This means we cannot directly relate its value to the values in our data set. Therefore, the variance appears somewhat arbitrary.

3.2.6 Standard deviation

Standard deviation is a measure of the spread of scores within a set of data.

- Population deviation = calculated using the entire population data set
- Sample deviation = calculated using the values in the sample

Most of the time we calculate the sample deviation and use it to estimate the population deviation.

$$\text{sample standard deviation} = \frac{\sqrt{\sum (X - \bar{X})^2}}{N - 1}$$

Where $\bar{X}$ = sample mean, $\sum$ = sum of, X = data piece (score), N = number of data pieces (scores) in sample.

The standard deviation is used in conjunction with the mean to summarise numerical data, not categorical data. In addition, the standard deviation, like the mean, is normally only appropriate when the data is not significantly skewed or has outliers.

3.2.7 Using R studio to calculate measures of spread

summary(dataset) will give minimum, lower quartile, median, mean, upper quartile, maximum

The command **range** only gives the minimum and maximum values of the dataset to calculate range type **diff(range(dataset))**

IQR(WWWusage) gives the interquartile range

sd(WWWusage) gives the standard deviation

boxplot(dataset) will give a box and whisker graph

3.3 Describing Distributions

We can describe the overall pattern of a distribution by its **shape, measures of central tendency, spread and any unusual values**. Usual values are generally known as **outliers**.

Outliers are individual values that fall outside the overall pattern.

3.3.1 Shape of distribution of data

The **mode** of a distribution is seen as the peak point of **visual** shape of the distribution.

Unimodal data (one mode) has one single peak.

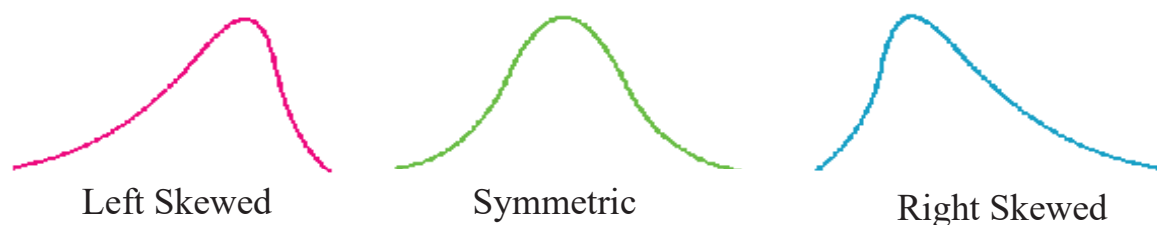
Bimodal data (two modes) has two peaks.

Multimodal (multiple modes) has more than two peaks.

In addition to the modes, you should state whether the distribution is symmetric or skewed.

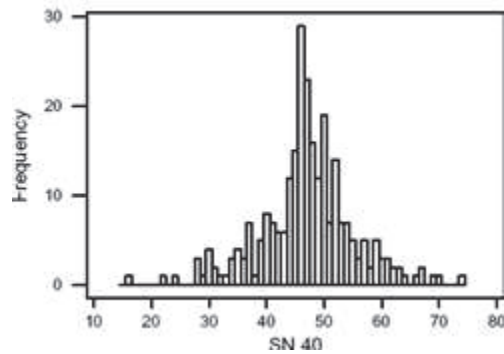
A distribution is **symmetric** if the right and left sides of the visual shape are approximately mirror images of each other.

A distribution is **skewed to the right** if the right side of the visual shape extends much farther out than the left side. It is **skewed to the left** if the left side of the visual shape extends much farther out than the right side.

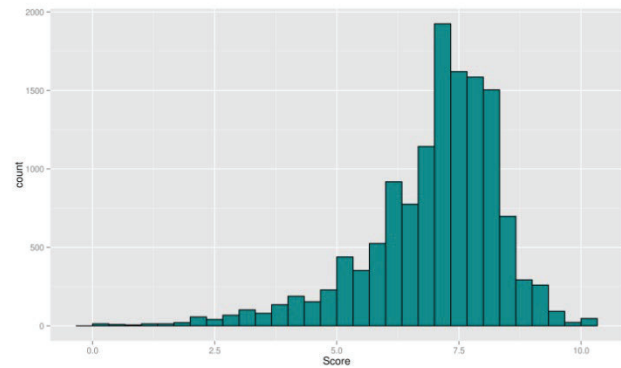
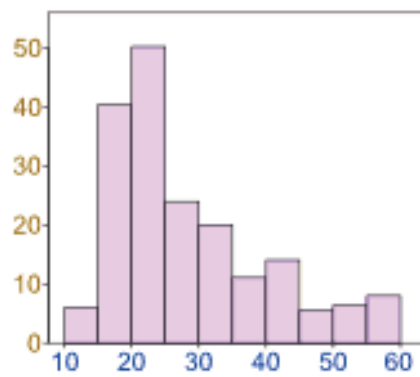


SOME EXAMPLES OF COMMON DISTRIBUTION SHAPES

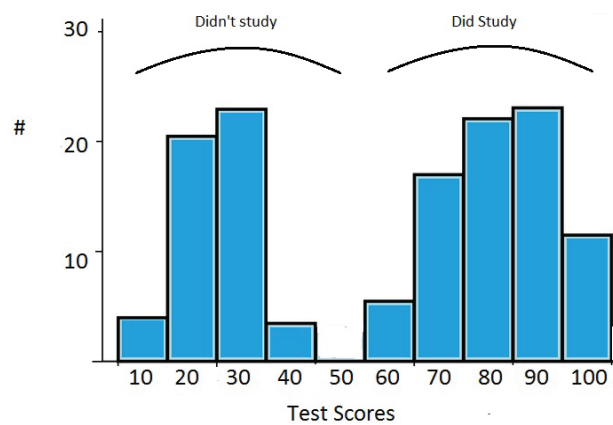
Symmetric, unimodal distribution with potential outliers



Right skewed, unimodal distribution



Bimodal distribution

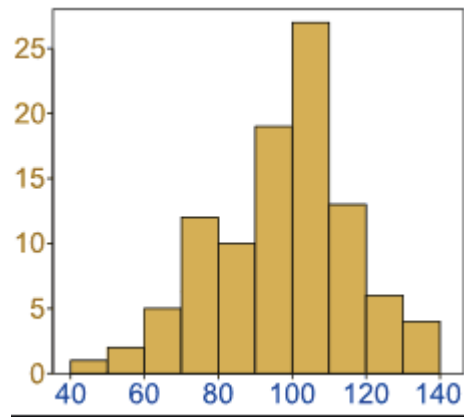


3.4 Visual representation of data

3.4.1 Histogram

A histogram is a graphical display of data using bars of different heights. In a histogram, each bar groups numbers into ranges. Taller bars show that more data falls in that range.

A histogram displays the shape and spread of numerical data.



3.4.2 Box and Whisker graphs

See 3.2.4

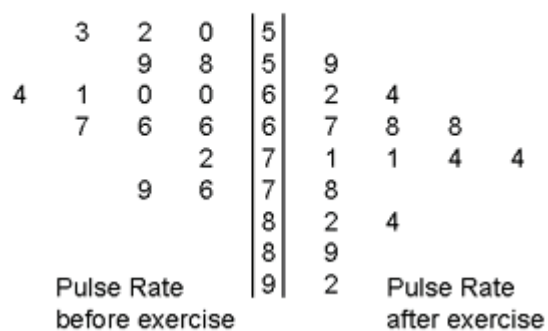
3.4.3 Stem and Leaf plots

A stem and leaf plot is a graph for displaying the distribution of numerical data. It is similar to a histogram but retains some information about individual values.

Ideally the numbers in the 'stem' represent the highest place-value digit in the values and the 'leaves' display the second highest place-value digits in each individual value.

To compare the distribution of a numerical variable for two categories, a back-to-back stem-and-leaf plot can be drawn, in which the stem is placed at the centre and the leaves for the values of the numerical variable for one category are drawn on one side of the stem and the leaves for the other category are drawn on the other side. Stem-and-leaf plots are particularly useful when the number of values to be plotted is not large.

Stem & Leaf Plot



Exercise: List the values (beats/minute) represented in this stem and leaf plot.

Chapter 4

Probability Distributions

4.1 Binomial Distribution

Reading: Tan p501–509 or LGR p467 – 472

4.1.1 Binomial experiment

A binomial experiment must satisfy the following conditions:

The same experiment is repeated a fixed number of times.
There are only two possible outcomes: success and failure.
The repeated trials are independent, so that the probability of success remains the same for each trial.

Example 1. The following experiments are binomial experiments:

Tossing a coin 7 times. A head is a success and a tail a failure.

Rolling a die 4 times. A one is a success and any other number a failure.

A machine is producing 75 items in a run. A working item is a success and a defective item a failure.

Example 2. A fair die is rolled 5 times. A six is a success and any other number a failure.

The probability of success in a single roll of the die is $p = \frac{1}{6}$

(a) Find the probability of rolling exactly 5 sixes.

- (b) Find the probability of rolling exactly 1 six.

4.1.2 Binomial probability

If p is the probability of success in a single trial of a binomial experiment,

the probability of x successes and $n - x$ failures in n independent repeated trials of the experiment is called the binomial probability. It is given by

$$P(x \text{ success}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Note that the notation $\binom{n}{x}$ is used for the number of combinations of x objects chosen from n distinct items. $\binom{n}{x}$ is expressed as $C(n,x)$ in chapter 2 .

Example 3. For a certain production line, the probability that any one item is defective is 0.01.

- a) Find the probability that there is exactly 1 defective item in a sample of 15 items from the production line.

- b) Find the probability that there is at most 1 defective item in a sample of 15 items from the production line.

Exercises:

1. A company gives prospective workers a 6-question, multiple-choice test. Each question has 5 possible answers, so that there is a 20% chance of answering the question correctly just by guessing. Find the probabilities for getting the following results for the test:
 - a) No correct answers
 - b) Exactly two correct answers

4.2 Normal Distribution

4.2.1 Normal Distribution

Reading: <https://www.analyticsvidhya.com/blog/2020/04/statistics-data-science-normal-distribution/>

This is one of the most important examples of a continuous probability distribution. It is the classic “bell curve”. Normal distributions often represent measurements found in nature.

Unlike the binomial distribution that deals with discrete values, the normal distribution allows for a continuous set of values. We can measure things like time, mass, length, etc.

There are two parameters for the Normal Distribution:

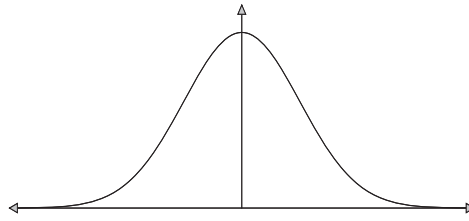
The Mean (μ) and the Standard Deviation (σ)

The mean tells us where to place the curve on a scale and the Standard Deviation tells us about the shape of the ‘bell’ curve.

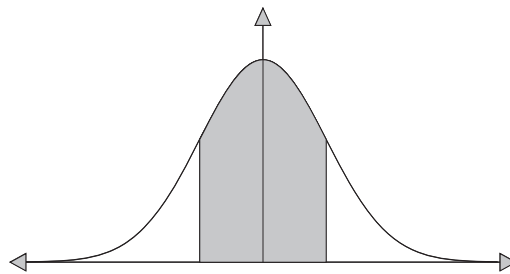
Most measurements in a set of data are near the middle. Normally distributed data is described by giving

- the mean (middle value)
- the standard deviation (spread)

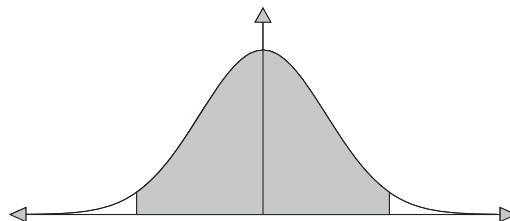
The curve of this data fits a bell shape.



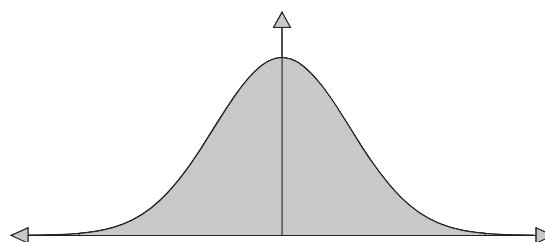
Note that 68% of the data is within 1 standard deviation either side of the mean. Data is likely to be in this region.



95% of the data is within 2 standard deviations either side of the mean. Data is very likely to be in this region

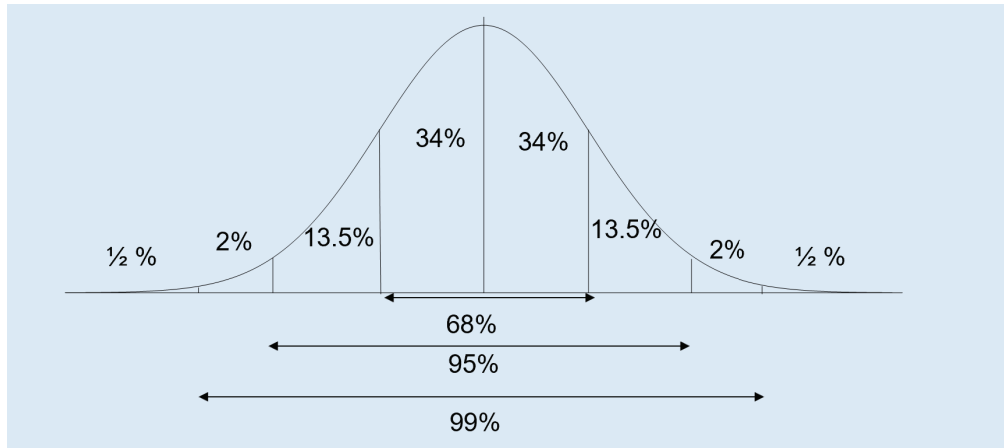


99% of the data is within 3 standard deviations either side of the mean. Data is almost certain to be in this region.



The normal curve is symmetrical and total probability adds to 100% or 1.

In summary:



Data is likely or probably in the range of 1 standard deviation

Data is very likely or very probably in the range of 2 standard deviations

Data is almost certain to be in the range of 3 standard deviations.

Example 1. The heights of a population of young women are normally distributed with a mean of 167 cm and standard deviation 6 cm. Calculate the probability that a student chosen at random has a height,

- a) between 161cm and 173cm. $P(161 < X < 173)$
- b) between 155cm and 185cm. $P(155 < X < 185)$
- c) less than 167cm. $P(X < 167)$
- d) greater than 161cm. $P(X > 161)$
- e) between 149cm and 173cm.

- f) greater than 173

4.2.2 Inverse Normal distribution problems

So far we have looked at problems where we have calculated probabilities. In inverse normal distribution problems, the probability, the mean and the standard deviation are given and we need to find the “x” value.

Example 1. For a dataset containing the heights of young women, the mean is 167 cm and the standard deviation 6 cm.

- a) Find below which height does 70% of the population fall?

- b) Find above which height is 25% of the population?

Exercises:

1. Suppose that the population of possible cholesterol measurements if you are healthy has a mean of 190 and a standard deviation of 10.
 - a) Further, suppose you know you should get concerned if your measurement ever gets up to the 97th percentile. What level of cholesterol does that represent?
 - b) If a random sample of size 50 is taken at the local community clinic, how many people will have a cholesterol measurement greater than 210?
2. Suppose you record how long it takes you to travel to work or university over many months and discover that the times are approximately bell-shaped with a mean of 23 minutes and a standard deviation of 2 minutes. How much time should you allow to make sure you are on time, 90% of the time?
3. The water roller coaster has a minimum height requirement of 1.52m. The average height of children who visit this amusement park is 1.22m with a standard deviation of 0.61m. Approximately what percentage of the children who come to the park qualify to ride the roller coaster. Assume that the distribution of children's heights is approximately normal.

4.3 Hypothesis testing

4.3.1 Populations and samples

Hypothesis testing is a method for making inferences or drawing statistical conclusions about a population parameter by using information for a sample drawn from the population.

In this course hypothesis testing is done for:

- the mean μ of a normal distribution,
- a population proportion p .

The first steps to follow are:

Step 1. Determine whether the test refers to ' p ' or ' μ '.

Step 2. Write down the null and the alternative hypotheses.

Step 3. Determine whether the test is one-sided or two-sided.

Step 4. Decide on the significance level for the test. Usually, it will be given.

4.3.2 Hypothesis testing for a population proportion ' p '

Hypothesis testing for a population parameter is used when a proportion of the population has a certain property or holds a certain opinion.

A sample of n items or people are chosen randomly, and their opinions or properties are recorded. A success in the sample is a person that agrees with the opinion, or an item that has the property. The number of successes r in the sample is distributed according to a binomial distribution with parameters n and p .

For testing a null hypothesis for a population proportion there are two methods:

- the p-value
- the critical region or rejection region

Example:

An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. Three people say they do.

Null hypothesis: (in words and in symbols)

H_0 : 40% of the residents in the town support the candidate

$$H_0: p = 0.4$$

Alternative hypothesis: (in words and in symbols)

H_a : Less than 40% of the residents in the town support the candidate

$$H_a: p < 0.4$$

This is a one-tailed test (left tail) at a 5% significance level

Test statistic: The number of people in a sample that supports the candidate.

Observation: In a sample of 20 people, 3 people supported the candidate.

Method 1: Find the p-value.

The 'p-value' is the probability of obtaining results at least as extreme as the observation, given that the null hypothesis is true. In this example, the p-value is the probability of obtaining three supporters or fewer, given that 40% of the all the residents support her ($p = 0.40$).

$$\text{p-value} = P(3 \text{ supporters or less out of } 20 | \text{proportion of supporters } p = 0.4)$$

Use the binomial distribution with $n = 20$ and $p = 0.4$ ($X \sim B(20, 0.4)$):

$$\text{p-value} = P(0 \text{ successes}) + P(1 \text{ success}) + P(2 \text{ successes}) + P(3 \text{ successes}) = 0.016$$

$$\text{p-value} < 0.05 \text{ (significance level)}$$

Conclusion:

As $p\text{-value} < 5\%$, we reject the null hypothesis. The candidate is over-estimating her support.

Method 2: Find the critical region or rejection region.

Find the largest number of successes out of 20 trials so that the cumulative probability remains less than 5%.

In formal notation: $P(\text{at most } x \text{ successes out of } 20) < 0.05$

Use the binomial distribution with $n = 20$ and $p = 0.4$:

$$P(x \leq 3) = 0.016 = 1.6\% < 5\%.$$

$$P(x \leq 4) = 0.51 = 5.1\% > 5\%.$$

The critical region is 3 or fewer supporters out of 20 people.

Conclusion:

As 3 people in the sample of 20 people support the candidate, we reject the null hypothesis.

The candidate is over-estimating her support.

Note that both the methods provide the same outcome.

The table below contains a summary of how to use the two methods.

One-sided test	Two-sided test
Null hypothesis: $H_0: p = p_0$	Null hypothesis: $H_0: p = p_0$
Alternative hypothesis: $H_a: p < p_0$ or $H_a: p > p_0$	Alternative hypothesis: $H_a: p \neq p_0$

p-value for left-tailed test: p-value = $P(r \text{ or fewer successes in the sample })$ p-value for right-tailed test: p-value = $P(r \text{ or more successes in the sample })$ Compare the p-value with 5%. If p-value > 0.05, reject H_0. If p-value < 0.05, do not reject H_0.	p-value: p-value = $2 \times P(r \text{ or fewer successes in the sample })$ Compare the p-value with 5%. If p-value > 0.05, reject H_0. If p-value < 0.05, do not reject H_0.
Critical region for left-tailed test: Use the binomial distribution with n trials and probability p for a success in a single trial. Find the largest number of successes so that the cumulative probability remains less than 5%. This is the critical value. If <i>observed value</i> $<$ <i>critical value</i>, reject H_0. If <i>observed value</i> $>$ <i>critical value</i>, do not reject H_0. Use a similar method for a right-tailed test.	Critical region: Use the binomial distribution with n trials and probability p for a success in a single trial. For a two-sided test we need two critical values: The largest number of successes so that the cumulative probability remains less than 2.5%. This is the lower critical value for the left tail of the distribution. The smallest number of successes so that $1 - \text{cumulative probability}$ remains less than 2.5%. This is the higher critical value for the right tail of the distribution. If the observed value is less than the lower critical value, or more than the higher critical value, reject H_0. Otherwise, do not reject H_0.
Interpret the result in terms of the given situation.	Interpret the result in terms of the given situation.

4.3.3 Hypothesis testing for a population mean ' μ '

A given random variable is distributed normally with mean μ and standard deviation σ .

A sample with size n is drawn from the population and the observed sample mean is $\bar{x}$.

The sample mean $\bar{x}$ is also a random variable, as different samples have different $\bar{x}$ -values.

The sample mean is normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.

The observed sample mean $\bar{x}$ is used to test a null hypothesis about a specific value μ_0 for the population mean μ .

We will discuss three methods for testing the null hypothesis for the mean μ of a normal distribution:

- the p-value
- the critical region or rejection region
- the z -score

This example is from Myimaths: Hypothesis testing 3

Example

Marks in an exam are normally distributed with $\mu = 56.9$ and $\sigma = 6.9$. A teacher looks at the marks of 11 of her students. Their mean mark is 56. She decides to test if the mean for the exam is lower than 56.9 at a 5% significance level.

Null hypothesis: (in words and in symbols)

H_0 : The mean mark for the test is 56.9

$H_0: \mu = 56.9$

Alternative hypothesis: (in words and in symbols)

H_a : The mean mark for the test is less than 56.9

$H_a: \mu < 56.9$

This is a one-tailed test (left tail) at a 5% significance level

Test statistic: $\bar{x}$ has a normal distribution with mean $\mu = 56.9$ and $sd = \frac{6.9}{\sqrt{11}} = 2.08$.

Observation: $\bar{x} = 56$ for a specific sample of 11 students.

Method 1: Find the p-value.

The 'p-value' is the probability of obtaining test results at least as extreme as the observation, given that the null hypothesis is true.

In this example, the p-value is the probability, given that $\mu = 56.9$ is true, that the sample mean $\bar{x}$ will be less than the observed value of 56.

p-value = $P(\bar{x} < 56)$ for $X \sim N(2.08, 56.9) = 0.3326$.

p-value = $0.3326 > 0.05$. (The significance level is 5%.)

Conclusion:

There is no evidence to reject the null hypothesis.

There is no evidence that the mean of the exam is less than 56.9.

Method 2: Find the critical region or rejection region.

Find the highest value for the sample mean $\bar{x}$ that will give a left tail of 0.05. This is called the critical value.

Critical region: $P(\bar{x} < \text{critical value}) = 0.05$.

Use the inverse normal distribution: critical value = 53.4787.

Rejection region is $\bar{x} < 53.4787$, and the observation 56 is outside the rejection region.

Conclusion:

There is no evidence to reject the null hypothesis.

There is no evidence that the mean of the exam is less than 56.9.

Method 3: Find the z-score.

For a standard normal distribution, the critical value is well-known.

For $Z \sim N(0,1)$ the critical z-value for a confidence level of 5% is -1.6449 as

$$P(Z < -1.6449) < 0.05.$$

Now find the z-score for the observation: $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{56 - 56.9}{2.08} = -0.4327$.

As $-0.4327 > -1.6449$, the observation is larger than the critical z-value.

Conclusion:

There is no evidence to reject the null hypothesis.

There is no evidence that the mean of the exam is less than 56.9.

Note that each of the three methods provides the same outcome.

The table below contains a summary of how to use the three methods for a significance level of 5%.

Test statistic: $\bar{x}$ has a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$. Use the observed value for $\bar{x}$ in the hypothesis testing.

One-sided test	Two-sided test
Null hypothesis: $H_0: \mu = \mu_0$	Null hypothesis: $H_0: \mu = \mu_0$
Alternative hypothesis: $H_a: \mu < \mu_0$ or $H_a: \mu > \mu_0$	Alternative hypothesis: $H_a: \mu \neq \mu_0$
p-value: p-value = $P(X < \bar{x})$ or p-value = $P(X > \bar{x})$ Compare the p-value with 5%. If p-value > 0.05, reject H_0 . If p-value < 0.05, do not reject H_0 .	p-value: p-value = $2 \times P(X < \bar{x})$ or p-value = $2 \times P(X > \bar{x})$ Compare the p-value with 5%. If p-value > 0.05, reject H_0 . If p-value < 0.05, do not reject H_0 .
Critical region for left-tailed test: $P(\bar{x} < \text{critical value}) = 0.05$. Use the inverse normal distribution to find the critical value. Rejection region: $\bar{x} < \text{critical value}$ If <i>observation</i> < <i>critical value</i> , reject H_0 . If <i>observation</i> > <i>critical value</i> , do not reject H_0 .	Critical region: $P(\bar{x} < \text{critical value}) = 0.025$ or $P(\bar{x} > \text{critical value}) = 0.025$ Use the inverse normal distribution to find the critical value. Rejection region: $\bar{x} < \text{critical value}$ or $\bar{x} > \text{critical value}$ If the observation is in the rejection region, reject H_0 .

Similar method for right-tailed test.	Otherwise, do not reject H_0 .
z-score for left-tailed test: For $Z \sim N(0,1)$ the critical value for a 5% level of significance is -1.6449. Find z-score for the observation: $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ If $z < -1.6440$, reject H_0. If $z > -1.6449$, do not reject H_0. Similar method for right-tailed test.	z-score: For $Z \sim N(0,1)$ the critical values for a 5% level of significance is -1.96 and 1.96. Find z-score for the observation: $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ If $z < -1.96$ or $z > 1.96$ reject H_0. Otherwise, do not reject H_0.
Interpret the result in terms of the given situation.	Interpret the result in terms of the given situation.

Chapter 5

Relations and functions

5.1 Linear functions

Reading: [T] p2 – 17 (on Course Resources) or [LGR] p2 – 15

This is a self study topic. Online material is available at:

www.mymaths.com → *Classis MyiMaths* → *algebra* → *graphs*

A linear function of the variable x can be written in the form

$$y = mx + c$$

or

$$f(x) = mx + c.$$

The graph of the function is a straight line.

The constant m gives the *slope* or *gradient* of the graph and the constant c the *intercept* on the y -axis.

You self-study should include the following topics:

- ordered pairs and the Cartesian coordinate system
- the equation for a line
- x -intercept and y -intercept of a line
- finding the slope of a line
- parallel and perpendicular lines
- examples of linear models

5.2 Power sets

[GT] Discrete mathematics: proofs, structures, and applications, Rowan Garnier and John Taylor (ebook on Course Resources)

Reading: [GT] Section 3.6, p114 - 118

The **power set** of a set A , denoted $\mathcal{P}(A)$, is the set of all subsets of A :

$$\mathcal{P}(A) = \{B : B \subseteq A\}.$$

Note that $A \in \mathcal{P}(A)$ and $\emptyset \in \mathcal{P}(A)$. Therefore $\mathcal{P}(A) \neq \emptyset$.

Example 1

Let $A = \{1, 2, 3\}$. Then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$$

Note that the elements of the power set are themselves sets. There are 8 subsets of A .

For a finite set A with k elements, there are 2^k elements in the power set $\mathcal{P}(A)$.

In terms of the notation for cardinality of sets:

$$\text{If } n(A) = k, \text{ then } n(\mathcal{P}(A)) = 2^k.$$

Example 2

Let $B = \{a, b, c, d\}$.

Then $n(B) = 4$ and $n(\mathcal{P}(B)) = 2^4 = 16$.

List the 16 elements of the power set $\mathcal{P}(B)$.

Exercises

1. List the 16 elements of the power set $\mathcal{P}(B)$ in Example 2.
2. If $A = \emptyset$, what is the power set $\mathcal{P}(A)$?
How many elements are there in the power set $\mathcal{P}(A)$?

5.3 Cartesian product of sets

Reading: [GT] Section 3.7, p122 -128

The **Cartesian product** $A \times B$ of two sets A and B is the set of all **ordered pairs** (a, b) where a belongs to A and b belongs to B :

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

If the sets A and B are finite sets, the **cardinality** (number of elements) of the Cartesian product $A \times B$ is

$$n(A \times B) = n(A) \times n(B).$$

Example 1

$A = \{1, 2, 3\}$ and $B = \{r, s\}$

Then $A \times B =$

The cardinality $n(A \times B) = n(A) \times n(B) =$

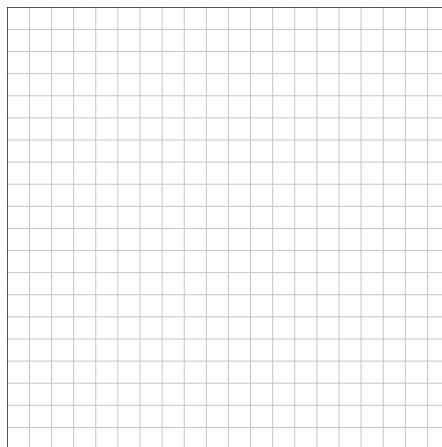
Example 2

$A = \{4, 5, 7\}$ and $B = \{2, 4, 6\}$

Then $A \times B =$

The cardinality $n(A \times B) = n(A) \times n(B) =$

Cartesian products of sets with numbers as elements can be presented in the Cartesian coordinate system.



Example 3

$M = \{\text{main courses served by a certain restaurant}\}$

$D = \{\text{desserts served by the same restaurant}\}$

Then $M \times D$ is all the two course meals that one can order at the restaurant (listing the main course first).

Example 4

$A = \{\text{cities of the world}\}$ and $B = \{\text{countries of the world}\}$

$(\text{Auckland, New Zealand}) \in A \times B$

$(\text{Auckland, Fiji}) \in A \times B$

$(\text{Fiji, Auckland}) \notin A \times B$ (Why?)

$(\text{Wanaka, India}) \notin A \times B$ (Why?)

Example 5

$\mathbb{R}$ is the set of real numbers.

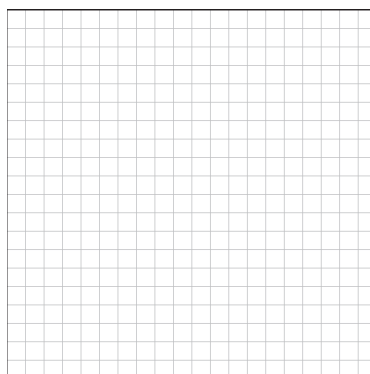
The following intervals of real numbers are given:

$A = [0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$ and $B = (0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$

Describe the Cartesian product $A \times B$. Show the Cartesian product $[0, 1] \times (0, 1)$ on the coordinate system.

The Cartesian product $A \times B = [0, 1] \times (0, 1) =$

Show the Cartesian product $A \times B = [0, 1] \times (0, 1)$ on the coordinate system.

**Exercises**

1. Let $P = \{-1, 0, 1\}$ and $Q = \{-1, 1\}$. Find $P \times Q$ and $n(P \times Q)$.
2. Show the Cartesian product $[0, 1) \times (0, 1]$ on a coordinate system.

5.4 Relations

Reading: [GT] 4.1 (p154 - 159), 4.2

Definition and notation

A **relation** R from a set A to a set B (or between A and B) is a subset of the Cartesian product $A \times B$.

If R is a relation from A to B , we say that $a \in A$ is related to $b \in B$ if $(a, b) \in R$.

We write $a R b$ for “ a is related to b ”.

We write $a \not R b$ for “ a is not related to b ”.

For a specific relation R the relationship between the entries in the ordered pair must be given.

Example 1

$A = \{\text{cities of the world}\}$

$B = \{\text{countries of the world}\}$

$R = \{(a, b) : a \text{ is the capital city of } b\}$

So $(\text{Wellington, New Zealand}) \in R$ and $(\text{Auckland, New Zealand}) \notin R$.

Also $a R b$ means that “ a is the capital city of b ”.

So $\text{Wellington } R \text{ New Zealand}$ and $\text{Auckland } \not R \text{ New Zealand}$.

If $A = B$, the relation R is from A to A and R is called a relation *on* A .

Example 2

Let $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) \in A \times A : a \leq b\}$.

We can list the elements of the relation R as a subset of $A \times A$:

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$

We can also write, for example: $1 R 1$, $1 R 2$, $2 R 3$ but $3 \not R 2$.

Representations of relations

The relation R on set A can be represented by a **directed graph**.

A directed graph is constructed using the following steps:

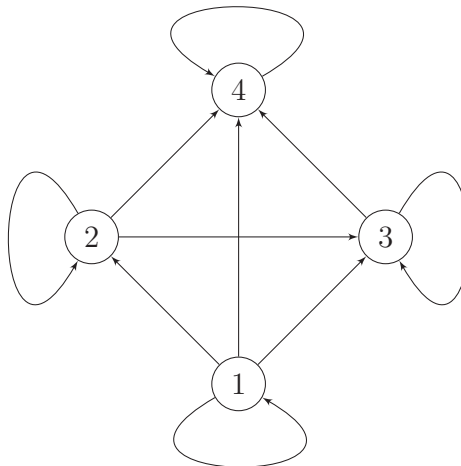
- Show each element of A as a point. These points are called **nodes**.
- Then draw an arrow from a to b if and only if $a R b$. These arrows are called **edges**.
- If $a R a$, draw an arrow that starts from a and returns to a .
- If $a R b$ and $b R a$, show the edge as a bi-directional arrow.

Example 2 (continued)

The directed graph for R is below. Note the directions of the arrows.

For example, $1 R 3$, and the arrow (edge) starts at the node for 1 and points towards the node for 3.

Also, $3 R 3$, and there is an arrow (edge) starting from the node for 3 and returning to the node for 3.



Properties of relations

Let R be a relation on set A . We say that R is:

- reflexive** if and only if $(a, a) \in R$ for every $a \in A$
- symmetric** if and only if $(a, b) \in R$ implies $(b, a) \in R$ for every a and b in A
- anti-symmetric** if and only if $(a, b) \in R$ and $(b, a) \in R$ implies $a = b$ for every a and b in A
- transitive** if and only if $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$ for every a, b and c in A

Example 2 (continued)

There are different ways to check whether the relation R has any of the four properties.

Option 1: List the ordered pairs in the set R and check the properties.

- *Reflexive:* The relation R in Example 2 *is reflexive* as $(1, 1)$, $(2, 2)$, $(3, 3)$, $(4, 4)$ are in R . The relation R is reflexive.
- *Symmetric:* If there is a single pair in R that does not satisfy this property, the relation is not symmetric. For instance, $(1, 2) \in R$, but $(2, 1) \notin R$. Therefore the relation R *is not symmetric*.
- *Anti-symmetric:* There are no pairs in (a, b) and (b, a) in R with $a \neq b$. R is *anti-symmetric*.
- *Transitive:* We have to consider all pairs in R that follows one another. For example, $(1, 2) \in R$ and $(2, 3) \in R$. For this pair it is true that $(1, 3) \in R$. Some other pairs to check are: $(1, 3)$ and $(3, 4)$; $(1, 2)$ and $(2, 4)$. Are there any other pairs that we need to check? The relation R is *transitive*.

Option 2: Provide a general argument for each of the four properties.

- *Reflexive:* A general argument is as follows: $(a, a) \in R$ for any $a \in A$ as $a \leq a$.
- *Symmetric:* If $(a, b) \in R$ it means that $a \leq b$. Choose a pair in R with $a \neq b$. Then $a < b$ and $(b, a) \notin R$. Therefore R is not symmetric.
- *Anti-symmetric:* If $(a, b) \in R$ and $(b, a) \in R$ it means that $a \leq b$ and $b \leq a$. Therefore $a = b$. The relation R *is anti-symmetric*.
- *Transitive:* A general argument is as follows: $(a, b) \in R$ and $(b, c) \in R$ means that $a \leq b$ and $b \leq c$. Therefore $a \leq c$, and $(a, c) \in R$. The relation R *is transitive*.

Option 3: Use the directed graph to check the four properties.

- *Reflexive:* At every node there is an edge starting at the node and returning to the node.
- *Symmetric:* All the edges connecting different nodes are all going in only one direction. All the edges must be bi-directional for a relation R that is symmetric.
- *Anti-symmetric:* Check that the graph has no bidirectional arrows connecting different points.
- *Transitive:* If three points are such that there are edges from the first to the second and from the second to the third, then there must also be an edge from the first to the third.

Exercises

1. Let $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) \in A \times A : a > b\}$.
 - List all the elements (ordered pairs) in R .
 - Construct a directed graph for R
 - Check whether R is reflexive, symmetric, anti-symmetric and/or transitive.
2. Let A be any set of living human beings. For a and b in A , let $a R b$ if a is a brother of b . Is R reflexive, symmetric, anti-symmetric and/or transitive?

5.5 Functions

Reading: [GT] 5.1 (p220 - 229)

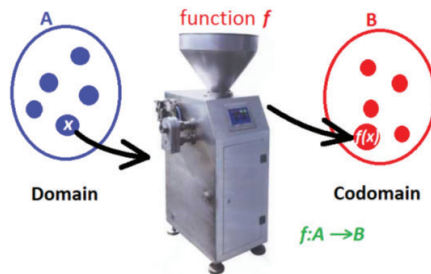
Definition and notation

A **function** f from a set A to a set B , written $f : A \rightarrow B$, is a rule which associates to each $a \in A$ a unique element $f(a) \in B$.

This definition of a function f is visualised as a “input-output” machine.

For an input $a \in A$ the function f produces the output $f(a) \in B$. The output $f(a)$ is also called the image of a .

The set A is called the *domain* of f , and the set B the *co-domain* of f .



This is a general definition and is used even when the sets A and B are non-numerical.

A function f is also a special kind of *relation*. We rewrite the definition in terms of ordered pairs in the Cartesian product $A \times B$.

A **function** f from a set A to a set B is a subset of $A \times B$ which satisfies the following condition:
For every (input) $a \in A$ there is exactly one (output) $b \in B$ for which $(a, b) \in f$.

The **image (or range)** of a function $f : A \rightarrow B$ is the set

$$\{b \in B : b = f(a) \text{ for some } a \in A\}.$$

Example 1

The function on the real numbers $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 4x^2$ has domain of the function f is $\mathbb{R}$.

For instance, $f(2) = 4 \times 2^2 = 16$ and $f(-2) = 4 \times (-2)^2 = 16$.

The codomain is $\mathbb{R}$.

The range of the function f is $\{y \in \mathbb{R} : y \geq 0\}$ as all the images are non-negative.

Example 2

Let A be the set of countries in the world.

Let B be the set of cities in the world.

Define the function $capital : A \rightarrow B$ by

$$capital(x) = y \text{ if and only if } y \text{ is the capital of country } x.$$

$capital$ is a function as every country x in the world has a capital city $capital(x)$.

The *domain* for $capital$ is the set of all countries in the world.

The *co-domain* for $capital$ is the set of all cities in the world.

The *range* for $capital$ is the set of all capital cities. The range is a subset of the co-domain.

A **function** f from a set A to a set B is

- *one-to-one* if no two elements of the domain have the same image;
- *onto* if every element of the codomain is the image of at least one element of the domain.

Example 3

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 4x - 3$ is one-to-one and onto.

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 4x^2$ is not one-to-one and not onto.

Exercises

1. What is the range of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 4x^2 - 1$?
2. Is the function $capital : A \rightarrow B$ in Example 2 one-to-one and/or onto?

Chapter 6

Matrices

6.1 Matrix definitions and operations

Reading: [T] p102 – 109, p115 – 121 or [LGR] p86 – 103

Matrix dimension

A *matrix* is a rectangular array of values.

It has *dimension* or *order* $m \times n$ if it contains m rows and n columns.

Note that the number of rows is specified first.

If $m = n$ it is called a *square* matrix.

If $m = 1$ it is called a *row* matrix.

If $n = 1$ it is called a *column* matrix.

Example 1

$$A = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix} \quad B = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \quad C = \begin{pmatrix} 7 \\ -3 \\ 2 \end{pmatrix}$$

A is a 2×3 matrix, B is a square 2×2 matrix, C is a column 3×1 matrix.

Notation

A general $m \times n$ matrix A is written as follows:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \text{ or } A = [a_{ij}].$$

Equal matrices

Two matrices A and B are said to be *equal*, denoted by $A = B$, if they have the same dimension, and the same corresponding entries, that is, the i, j -entry of A is the same as the i, j -entry of B .

Example 2

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \end{pmatrix}$$

$A \neq B$. Why?

Zero matrix

The *zero matrix* $O_{m \times n}$ is the $m \times n$ matrix all of whose entries are zero.

Example 3

$$O_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad O_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Diagonal matrix

A *diagonal matrix* is a square matrix $[a_{ij}]$ for which $a_{ij} = 0$ if $i \neq j$, that is, all the entries that do not lie on the diagonal from the top left to the bottom right of the matrix must be zero.

Example 4

$$A = \begin{pmatrix} -3 & 0 \\ 0 & 5 \end{pmatrix} \qquad B = \begin{pmatrix} -0.5 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

Identity matrix

The *identity matrix* I_n is a diagonal $n \times n$ matrix for which all the diagonal entries are 1.

Example 5

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Transpose

The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^T obtained by interchanging the rows and columns of A , so the i, j -entry of A becomes the j, i -entry of A^T .

Example 6

$$P = \begin{pmatrix} 4 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix} \quad P^T = \begin{pmatrix} & \\ & \end{pmatrix}.$$

Matrix addition

For two matrices to be added together they must have the same dimension.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are both $m \times n$ matrices then their *sum* is the $m \times n$ matrix $A + B = [a_{ij} + b_{ij}]$. Every i, j -entry of A is added to the i, j -entry of B .

Example 7

$$A + B = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix} + \begin{pmatrix} 3 & 0 & -4 \\ \frac{1}{3} & 9 & -1 \end{pmatrix} = \begin{pmatrix} & & \\ & & \end{pmatrix}.$$

Scalar multiplication

The *scalar multiple* of an $m \times n$ matrix $A = [a_{ij}]$ with a number k is the matrix $kA = [ka_{ij}]$. Every entry of the matrix is multiplied by the number.

Example 8

$$2A = 2 \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 24 & 14 & -2 \\ 4 & 0 & 3 \end{pmatrix}.$$

Matrix multiplication

For two matrices to be multiplied together the first must have the same number of columns as the second has rows.

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix then their *product* is an $m \times p$ matrix.

$AB = [a_{i1}b_{1j} + \cdots + a_{in}b_{nj}]$, that is, to obtain the i, j -entry of the product AB one must multiply entries along row i of A with the entries down column j of B and add together the n products:

$$AB = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{i1} & \cdots & a_{in} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} c_{11} & \cdots & c_{1p} \\ \vdots & & \vdots \\ c_{i1} & \boxed{c_{ij}} & \vdots \\ \vdots & & \vdots \\ c_{m1} & \cdots & c_{mp} \end{pmatrix}$$

where $c_{ij} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj}$.

Example 9

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}$$

A is a 2×2 matrix and B is a 2×2 matrix. The product AB will be a 2×2 matrix.

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} =$$

Note that usually $AB \neq BA$.

$$BA = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} =$$

Example 10

$$A = \begin{pmatrix} 12 & 7 & -1 \\ 2 & 0 & -3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -1 & 0 & 1 \\ 2 & 0 & 0 & 1 \\ 6 & -4 & 0 & 1 \end{pmatrix}$$

Which of the following products are possible? AB , BA , AC , CA , BC , CB .

Calculate them.

Exercises

1. Find $(AB)C$ and $A(BC)$. Are the two products the same?

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} \quad C = \begin{pmatrix} -1 & 4 \\ 2 & -3 \end{pmatrix}$$

2. Find the following products: AC , CA and AB .

$$A = \begin{pmatrix} -2 & -3 & 7 \\ 1 & 5 & 6 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad C = \begin{pmatrix} -1 & 4 \\ 2 & -3 \\ 4 & 0 \end{pmatrix}$$

6.2 Inverse of a matrix

Reading: [LGR] p580 – 587, p107 – 111

A square $n \times n$ matrix A has an inverse if and only if there exists another $n \times n$ matrix B such that

$$AB = BA = I_n.$$

We call B the **inverse** of A and write $A^{-1} = B$.

Example 1

The matrix $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$ has the inverse $A^{-1} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$.

To show that this is true, calculate the following two matrix products:

$$\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Inverse of a 2×2 matrix

The following procedure is used to calculate the inverse of a 2×2 matrix A ,

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

First find the number $a_{11}a_{22} - a_{12}a_{21}$. This number is called the **determinant** $\det(A)$ of the matrix A .

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Exercises

- Find the inverses for the following matrices, if they exist:

(a) $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$

(b) $A = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$

6.3 Transformation with matrices

Reading: MyiMaths/Classic MyiMaths/IGCSE/Matrices

A point in the plane is represented by the column matrix $\begin{pmatrix} x \\ y \end{pmatrix}$.

Matrices can be used to change the position of the point.

Example 1

Let $P = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ represent a point on the x -axis. Let $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

Then $AP = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$.

The point P is reflected in the y -axis and AP is on the negative x -axis.

A geometric figure with straight line sides can be transformed by multiplying the vertices with matrices with specific properties.

The lessons on MyiMaths have examples of reflections, rotations, enlargements, stretches and shear of geometric figures.

Here is a summary of the most important transformation matrices.

Reflections

The following are **reflection matrices**:

- $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ reflects the point in the line $x = 0$ (y -axis)
- $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ reflects the point in the line $y = 0$ (x -axis)
- $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ reflects the point in the line $y = x$
- $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ reflects the point in the line $y = -x$

Rotations

Some rotations with centre of rotation at the origin can be represented with rotation matrices.

The following are **rotation matrices**:

- $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ rotates a point clockwise through 90°
- $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ rotates a point anticlockwise through 90°
- $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ rotates a point through 180°

Enlargements and stretches

The enlargement matrix $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$ (k a positive number) produces an enlargement with scale factor k .

The **stretch matrix** $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$ (k a positive number) is a stretch in the x -direction with scale factor k .

The **stretch matrix** $\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ (k a positive number) is a stretch in the y -direction with scale factor k .

Shear transformation

The **shear matrix** $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ produces a shear in the x -direction with shear factor k .

The **shear matrix** $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$ produces a shear in the y -direction with shear factor k .

Exercises

1. Show that the reflection matrices are their own inverses.
2. Find the determinant of each of the transformation matrices.
3. Find the inverse matrix for each of the transformation matrices.

6.4 Markov chains

Reading: [T] p498 – 503 or [LGR] p558 – 564

States and transitions

In a Markov process, at any particular **stage**, a system is in one of n different **states**.

Example 1

Urban-suburban population flow:

In a large city a person is staying either in the inner city (State 1) or in the suburbs (State 2). City planners record the the population flow from one year (stage) to the next.

Transition diagram

In a Markov process, there are transition probabilities associated with the transition from one state to the next. These probabilities are conditional probabilities.

Example 1 (continued)

For a particular city it is predicted that each year 3% of people staying in the inner city will move to the suburbs and 6% of people staying in the suburbs will move to the inner city.

Sketch a transition diagram for this process if S_1 means that a person stays in the inner city and S_2 that a person stays in the suburbs.

The same information can be presented in a table:

		This year	
		Inner city	Suburbs
Next year	Inner city	0.97	0.06
	Suburbs	0.03	0.94

Transition matrix

A transition matrix contains the transition probabilities.

If P is the transition matrix, the element p_{ij} is the conditional probability that the next state will be S_i given that the current state is S_j .

This means that the present states are presented by the columns of the matrix and the new states by the rows.

Example 1 (continued)

Find the transition matrix for the urban-suburban population flow process.

State 1 is living in the inner city.

State 2 is living in the suburbs.

$$P = \begin{pmatrix} 0.97 & 0.06 \\ 0.03 & 0.94 \end{pmatrix}$$

Note that $p_{11} = P(\text{inner city next year}|\text{inner city this year}) = 0.97$.

$p_{21} = P(\text{suburbs next year}|\text{inner city this year}) = 0.03$

$p_{12} = P(\text{inner city next year}|\text{suburbs this year}) = 0.06$.

$p_{22} = P(\text{suburbs next year}|\text{suburbs this year}) = 0.94$.

Properties of a transition matrix

A transition matrix P has the following properties:

- It is a square matrix.

Reason: All possible states are used both as rows and as columns.

- All entries in the matrix are between 0 and 1, inclusive.

Reason: The entries are probabilities.

- The sum of entries in each column must be 1.

Reason: The entries in a column give the probabilities for changing from the column state to one of the row states.

Markov chain

A **Markov chain** is a collection of states $S_1, S_2, \dots, S_n$ together with a transition matrix P .

The entry p_{ij} in the matrix P is the probability of the transition $S_j \rightarrow S_i$.

Properties of a Markov chain

A sequence of trails of an experiment is a Markov chain if

- the outcome of each experiment is one of a discrete set of states,
- the outcome of each experiment depends only on the present state, and not on any past states,
- the probabilities remain constant from one transition to the next.

Distribution of states

A **probability vector** or **distribution vector** is a column matrix with nonnegative entries that add up to 1.

If a Markov chain had initial probability vector or distribution vector X_0 and transition matrix P , then the probability vector after one repetition (or at the next stage) X_1 will be given by a matrix product

$$X_1 = PX_0.$$

After two repetitions, the probability vector X_2 will be

$$X_2 = PX_1 = P(PX_0) = PPX_0 = P^2X_0.$$

After k repetitions, the probability vector X_k will be

$$X_k = P^k X_0.$$

Example 1 (continued)

At present, 65% of the population stays in the inner city and 35% in the suburbs. If the total population remains constant, what will the population distribution be 1 year from now? What will the population distribution be 2 years from now?

Exercises

1. Assume that at present 35% of the population stays in the inner city and 65% in the suburbs. Use the same transition probabilities as in Example 1. If the total population remains constant, what will the population distribution be 1 year from now? What will the population distribution be 2 years from now?

6.5 Regular Markov chains

Reading: [T] p508 – 513 or [LGR] p569 – 575

Regular transition matrices

A transition matrix P is **regular** if some power of the matrix contains all positive entries (no zero entries).

A Markov chain is a **regular Markov chain** if its transition matrix is regular.

Example 1

Decide whether the following transition matrices are regular.

First check whether the matrices are transition matrices.

$$(a) P = \begin{pmatrix} 0.75 & 0 & 0.6 \\ 0.25 & 0.5 & 0.4 \\ 0 & 0.5 & 0 \end{pmatrix}$$

$$(b) P = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{pmatrix}$$

If the transition matrix P contains some zero entries, and P^2 contains zeroes in the same positions, then these zeroes will appear in all higher powers of P . This means that P is not regular.

We may use the transition diagram of the Markov chain to test whether the transition matrix is regular. It should be possible to go, in some number of moves, from any state to any state.

Example 2

Use a transition diagram to decide whether the following transition matrices are regular.

$$(a) P = \begin{pmatrix} 0.75 & 0 & 0.6 \\ 0.25 & 0.5 & 0.4 \\ 0 & 0.5 & 0 \end{pmatrix}$$

$$(b) P = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{pmatrix}$$

Long-range predictions for regular Markov chains**Example 3**

For a particular city it is predicted that each year 3% of people living in the inner city will move to the suburbs and 6% of people living in the suburbs will move to the inner city.

If State 1 is living in the inner city and State 2 is living in the suburbs, the transition matrix is given by

$$P = \begin{pmatrix} 0.97 & 0.06 \\ 0.03 & 0.94 \end{pmatrix}$$

Assume that the initial distribution vector is $X_0 = \begin{pmatrix} 0.65 \\ 0.35 \end{pmatrix}$.

Then PX_0 is the distribution vector after one year.

P^2X_0 is the distribution vector after 2 years.

P^kX_0 is the distribution vector after k years.

A long-term prediction for the population distribution can be calculated.

In the table the long-term predictions are calculated for three different initial distribution vectors.

X_0	$X_1 = PX_0$	$X_2 = P^2X_0$	$X_3 = P^3X_0$	$X_{20} = P^{20}X_0$	$X_{50} = P^{50}X_0$
$\begin{pmatrix} 0.65 \\ 0.35 \end{pmatrix}$	$\begin{pmatrix} 0.651 \\ 0.349 \end{pmatrix}$	$\begin{pmatrix} 0.653 \\ 0.347 \end{pmatrix}$	$\begin{pmatrix} 0.654 \\ 0.346 \end{pmatrix}$	$\begin{pmatrix} 0.664 \\ 0.336 \end{pmatrix}$	$\begin{pmatrix} 0.667 \\ 0.333 \end{pmatrix}$
$\begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}$	$\begin{pmatrix} 0.697 \\ 0.303 \end{pmatrix}$	$\begin{pmatrix} 0.694 \\ 0.306 \end{pmatrix}$	$\begin{pmatrix} 0.692 \\ 0.308 \end{pmatrix}$	$\begin{pmatrix} 0.672 \\ 0.328 \end{pmatrix}$	$\begin{pmatrix} 0.667 \\ 0.333 \end{pmatrix}$
$\begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$	$\begin{pmatrix} 0.515 \\ 0.485 \end{pmatrix}$	$\begin{pmatrix} 0.529 \\ 0.471 \end{pmatrix}$	$\begin{pmatrix} 0.541 \\ 0.459 \end{pmatrix}$	$\begin{pmatrix} 0.641 \\ 0.359 \end{pmatrix}$	$\begin{pmatrix} 0.665 \\ 0.335 \end{pmatrix}$

The long-term prediction of a Markov chain does not depend on the initial distribution vector X_0 .

Equilibrium vector of a regular Markov chain

For a regular Markov chain with transition matrix P , there is a unique distribution vector V such that, for any distribution vector X_0

$$X_k = P^k X_0 \approx V$$

for large values of k .

The vector V is called the **equilibrium vector** or the **fixed vector** of the Markov chain.

Calculating the equilibrium vector

To find V by calculating $X_k = P^k X_0$ for large values of k as we did in the table, is a tedious calculation. We can find V by solving a system of linear equations.

If $X_k = P^k X_0 \approx V$, then $X_{k+1} = P^{k+1} X_0 \approx V$.

But $P^{k+1} X_0 = P(P^k X_0) \approx PV$.

The condition that we use to calculate V is

$$PV = V.$$

Example 4

Find the equilibrium vector for the regular transition matrix $P = \begin{pmatrix} 0.97 & 0.06 \\ 0.03 & 0.94 \end{pmatrix}$.

We solve the system of equations $PV = V$.

This gives the system of equations

$$\begin{aligned} 0.97v_1 + 0.06v_2 &= v_1 \\ 0.03v_1 + 0.94v_2 &= v_2 \end{aligned}$$

If we simplify these equations, we find

$$\begin{aligned} -0.03v_1 + 0.06v_2 &= 0 \\ 0.03v_1 - 0.06v_2 &= 0 \end{aligned}$$

The two equations are the same, and we require another condition in order to calculate V . Remember that V is a probability vector, and therefore $v_1 + v_2 = 1$.

The system of equations that we have to solve to find the equilibrium vector V is:

$$\begin{aligned} -3v_1 + 6v_2 &= 0 \\ v_1 + v_2 &= 1 \end{aligned}$$

Solving $AX = B$ using inverse matrix A^{-1}

A $n \times n$ system of linear equations is written in matrix form as:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix},$$

or

$$AX = B$$

Multiply both sides with the inverse matrix A^{-1} :

$$A^{-1}(AX) = A^{-1}B$$

Use the rules for matrix multiplication:

$$A^{-1}(AX) = (A^{-1}A)X = IX = X$$

The solution of the system is given by:

$$X = A^{-1}B$$

Example 5

Use an inverse matrix to solve the following system of linear equations

$$\begin{aligned}x + 3y &= 2 \\ 2x + 5y &= -3\end{aligned}$$

In matrix notation this system is given by

$$AX = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} = B$$

We know that $A^{-1} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$. (Check this!)

Use the inverse matrix A^{-1} to solve the system of equations:

$$X = A^{-1}B = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -19 \\ 7 \end{pmatrix}$$

Exercises

- Which of the following transition matrices are regular transition matrices?
 - $P = \begin{pmatrix} 0.2 & 0.9 \\ 0.8 & 0.1 \end{pmatrix}$
 - $P = \begin{pmatrix} 1 & 0.65 \\ 0 & 0.35 \end{pmatrix}$
 - $P = \begin{pmatrix} 0.4 & 0.7 \\ 0.6 & 0.3 \end{pmatrix}$
- Find the equilibrium vectors for the regular transition matrices in 1.

Hints for exercises

1.2 Sets

1. (a) $A = \{spring, summer, autumn, winter\}$
(b) $\{2, 4, 6\}$
2. (a) 6 subsets fit the conditions (list them)
(b) 15 subsets fit the conditions (list them)
(c) 8 subsets fit the conditions (list them)

1.3 Applications of Venn diagrams

2. 142 households

1.4 Probability and sets

2. $n(S) = 8$ and the probabilities for the events are $\frac{1}{8}$, $\frac{3}{8}$ and $\frac{1}{8}$.
3. The probabilities are
 - (a) $\frac{1}{4}$
 - (b) $\frac{1}{2}$
 - (c) $\frac{2}{13}$
 - (d) $\frac{3}{13}$
 - (e) $\frac{1}{13}$
 - (f) $\frac{4}{13}$
4. (a) not mutually exclusive
(b) not mutually exclusive
(c) mutually exclusive

1.5 Conditional probability

1. $\frac{11}{51}, \frac{4}{51}, \frac{25}{204}$
2. independent
3. 0.05, 0.4
4. 0.2, 0.4
5. 0.06, 0.85, 0.5275

2.1 Permutations

1. 1008, 2000
2. 156, 3.1×10^9 , n , $n!$, $n!$
3. 120
4. 78125, 10000, 544320
5. 456976, 16 times as many as before

2.2 Combinations

1. 792, 1.8×10^{12} , 1, n
3. 4620, 3432
4. 1716
5. 48, 658008, 844272

2.3 Counting principles and probability

1. 0.6501
2. (a) 8.417×10^{-8}
(b) 0.03846
(c) 0.6644
(d) 0.5417
3. (1) 0.6
(2) 0.4
4. 0.3824

4.1 Binomial distribution

1. (a) 0.2621
(b) 0.2458

4.2 Normal distribution

1. (a) Above 208.8
(b) 1 person
2. 25.6 minutes
3. 31%

5.2 Power sets

1. $B = \{a, b, c, d\}$

$$\mathcal{P}(B) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \\ \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, B\}$$

2. If $A = \emptyset$, then $n(A) = 0$ and $n(\mathcal{P}(A)) = 2^0 = 1$.

$\mathcal{P}(A) = \{\emptyset\}$ or A has only one subset.

In other words, the empty set $\emptyset$ is the only subset of the empty set $\emptyset$.

5.3 Cartesian products of sets

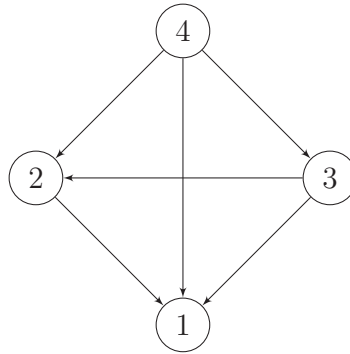
1. $P \times Q = \{(-1, -1), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 1)\}$

$$n(P \times Q) = n(P) \times n(Q) = 3 \times 2 = 6$$

Show the points on the Cartesian coordinate system to visualise the Cartesian product.

5.4 Relations

1.
 - $R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$
 -



- R is not reflexive and not symmetric. R is anti-symmetric and transitive.
- 2. R is not reflexive, not symmetric, not anti-symmetric and not transitive.

5.5 Functions

1. Range = $\{x \in \mathbb{R} | x \geq -1\}$
2. The function *capital* is one-to-one as each capital in B is associated with exactly one country.

The function *capital* is not onto, as there are some cities in B that are not capitals of any country.

6.1 Matrix definitions and operations

$$1. (AB)C = A(BC) = \begin{pmatrix} 19 & -11 \\ 32 & -8 \end{pmatrix}$$

$$2. AC = \begin{pmatrix} 24 & 1 \\ 33 & -11 \end{pmatrix}$$

$$CA = \begin{pmatrix} 6 & 23 & 17 \\ -7 & -21 & -4 \\ -8 & -12 & 28 \end{pmatrix}$$

$$AB = \begin{pmatrix} 13 \\ 29 \end{pmatrix}$$

6.2 Inverse of a matrix

$$1. \det(A) = 2 \text{ and } A^{-1} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{pmatrix}$$

2. $\det(A) = 1$ and $A^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$

6.3 Transformations with matrices

1. Show that the matrix product of a reflection matrix with itself gives the identity matrix $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
2. A reflection matrix has determinant -1 ; a rotation matrix and a shear matrix have determinant 1 ; an enlargement matrix has determinant k^2 ; a stretch matrix has determinant k .
3. Use the procedure in 6.3 to find the inverse matrix for each of the transformation matrices.

6.4 Markov chains

1. $X_0 = \begin{pmatrix} 0.35 \\ 0.65 \end{pmatrix}$
 $X_1 = \begin{pmatrix} 0.3785 \\ 0.6215 \end{pmatrix}$
 $X_2 = \begin{pmatrix} 0.4044 \\ 0.5956 \end{pmatrix}$

6.5 Regular Markov chains

1. (a) and (c) are regular transition matrices. For (b) it is impossible to reach state 1 from state 2, and the transition matrix is not regular.
2. (a) $V = \begin{pmatrix} \frac{9}{17} \\ \frac{8}{17} \end{pmatrix}$
(c) $V = \begin{pmatrix} \frac{7}{13} \\ \frac{6}{13} \end{pmatrix}$